PROJECT TECHNICAL DOCUMENTATION

# BioSpecInfo

*Web platform for computational chemistry, physics and mathematics*

# Open-Source Licensing & IP Compliance Matrix

## DOCUMENT CONTROL

| | |
|---|---|
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| **Author** | Samuele Pio Provenzano |
| **Language** | English |
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# Open-Source Licensing & IP Compliance Matrix — BioSpecInfo

| Field | Value |
|---|---|
| **Software** | BioSpecInfo |
| **Author / owner of proprietary code** | Samuele Pio Provenzano |

## 1. Third-party dependency matrix

All bundled third-party libraries are **open-source under permissive licenses** (MIT / BSD-3-Clause). **No viral copyleft dependency (GPL / AGPL / LGPL)** is used, so there are no obligations to redistribute the proprietary code.

| Component | Repo file | Use | License |
|---|---|---|---|
| **RDKit** (MinimalLib / rdkit-js) | `RDKit_minimal.js`, `RDKit_minimal.wasm` | Cheminformatics engine (SMILES/InChI/InChIKey/SMARTS, descriptors, fingerprints, 2D depiction) | **BSD-3-Clause** |
| **3Dmol.js** | `3Dmol-min.js` | 3D viewer for molecules/macromolecules (WebGL) | **BSD-3-Clause** |
| **three.js** | `three.min.js`, `three_bloom.js`, `gltf_loader.js` | WebGL 3D rendering (anatomy, astrochemistry, mechanisms) | **MIT** |
| **SmilesDrawer** | `smiles-drawer.min.js` | 2D depiction of SMILES strings | **MIT** |
| **sql.js** (SQLite compiled to WASM) | `lib/sql-wasm.js`, `lib/sql-wasm.wasm` | Client-side SQL playground | **MIT** |
| **PDF.js** | `vendor/pdfjs/` | Opening PDF documents in the spectrum reader: every page drawn and its text layer | **Apache-2.0** |

> The exact versions are those of the files included in the repository. Each permissive license requires preserving the original copyright notice, which must be kept in the library files or in a `THIRD_PARTY_NOTICES` file.

## 2. External services/APIs (not redistributed)

These services are queried over the network at runtime (their data is not part of the package); their use is subject to the respective terms.

| Service | Use |
|---|---|
| **PubChem PUG-REST** (NIH/NCBI) | IUPAC/common name, CAS, formula, 3D SDF models, GHS/CLP pictograms |
| **NIST WebBook · SDBS · MassBank · nmrshiftdb2 · SpectraBase** | Links to reference experimental spectra |
| **NASA / ESA** | Astronomical images and data (Astrochemistry section) |

## 3. Intellectual property of the proprietary code

- BioSpecInfo's application code (UI logic, the proprietary SVG mechanism engine,

the CIP stereochemistry viewer in Canvas, the heuristic spectral predictors, the educational content, the internal databases) is the **author's original work**, except for the third-party libraries listed above.

- **No copyleft-licensed code** is embedded in the proprietary source: the author

retains full licensing freedom over their own code.

### 3.1 Project license

The repository includes a **proprietary — "All rights reserved"** `LICENSE` file: the code is viewable and assessable (personal study and professional/academic evaluation), but **copying, modification, redistribution and commercial use require the Author's written permission**. This choice protects the project from unauthorised reuse while keeping it presentable to recruiters and companies. Third-party components remain governed by their permissive licenses.

## 4. Compliance summary

| Item | Result |
|---|---|
| Viral copyleft dependencies (GPL/AGPL) | **None** |
| Dependency licenses | Permissive (MIT / BSD-3-Clause) |
| Obligation to keep copyright notices | Yes (in library files / NOTICES) |
| Licensing freedom over proprietary code | Full |
| Project `LICENSE` file | ✅ Present — proprietary "All rights reserved" |